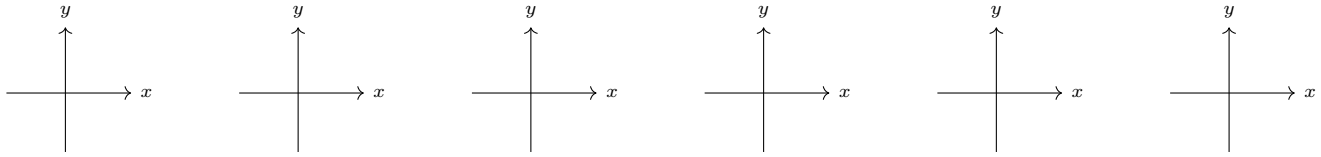
Lesson 1: Angles in Standard Position and Reference Angles

Assignment

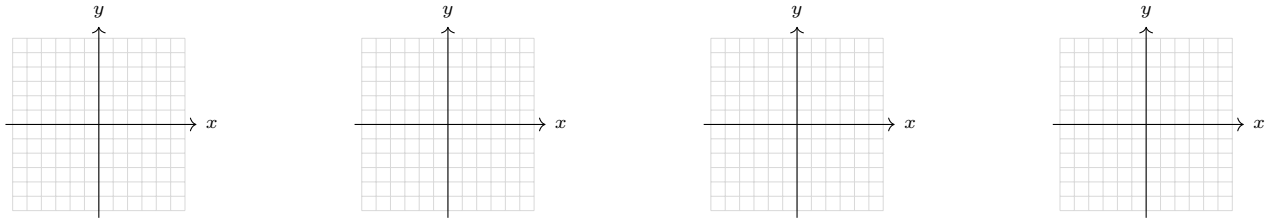
1. Sketch each angle in standard position and state the quadrant in which it terminates (or the axis it lies on).

(a) 140° (b) 215° (c) 285° (d) 25° (e) 180° (f) 270°



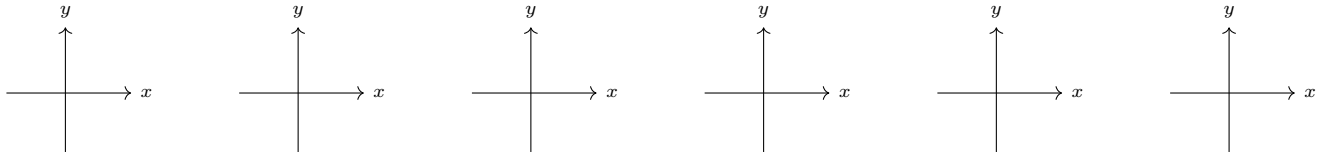
2. Each point is on the terminal arm of an angle θ in standard position. Plot the point, draw θ, and state the quadrant (or axis).

(a) $P(5,-2)$ (b) $Q(-3,6)$ (c) $R(-4,-1)$ (d) $S(0,-3)$



3. Sketch each angle and its reference angle. Determine θ_R.

(a) 236° (b) 302° (c) 99° (d) 64° (e) 165° (f) 360°



4. Without sketching, state the quadrant (or axis) and the reference angle.

(a) 344° (b) 101° (c) 258° (d) 12° (e) 189° (f) 271°

5. Sketch each rotation. State the angle from 0° to 360° with the same terminal arm.

(a) −45° (b) −200° (c) 390° (d) 600°



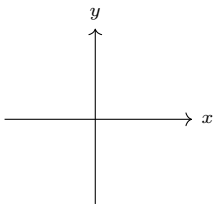
6. Determine the rotation angle θ, $0° \leq \theta \leq 360°$.

θ_R	Terminal arm	Sketch	θ
22°	quadrant 2		
45°	quadrant 3		
81°	quadrant 4		
13°	quadrant 1		

θ_R	Terminal arm	Sketch	θ
67°	quadrant 3		
5°	quadrant 2		
0°	negative x-axis		
90°	positive y-axis		

7. (a) On the axes, sketch a reference angle of 65° in each of the four quadrants.

(b) State the rotation angle in each quadrant.



8. Complete the table. Each row lists the four angles from 0° to 360° with the same reference angle.

Reference angle	Quadrant 1	Quadrant 2	Quadrant 3	Quadrant 4
18°				
63°				
		107°		
			222°	
				335°

9. Determine the three other angles from 0° to 360° with the same reference angle as the given angle.

(a) 147° (b) 316° (c) 203°

10. The point $P(2, 6)$ is on the terminal arm of an angle of 72° (to the nearest degree). For each reflection of P, state the coordinates of the image and the rotation angle whose terminal arm passes through it.

(a) in the y-axis (b) in the x-axis (c) in both axes

11. The point $P(a, b)$, with $a > 0$ and $b > 0$, is on the terminal arm of an angle θ. Write an expression in terms of θ for the rotation angle whose terminal arm passes through each point.

(a) $(-a, b)$ (b) $(-a, -b)$ (c) $(a, -b)$

12. The terminal arm of an angle θ passes through $(-5, 9)$, and $\theta \approx 119°$.

(a) What is the reference angle of θ?

(b) The point $(-5, 9)$ is reflected in the x-axis. State the image point and the angle from 0° to 360° whose terminal

arm passes through it.

13. Explain each student's error, then give the correct reference angle. A sketch will help.

(a) Jordan says the reference angle of 305° is 35°.

(b) Kim says the reference angle of 160° is 160°, "because it is measured from the positive x-axis."

14. An angle of 227° in standard position has a reference angle of

A. 43° **B.** 47° **C.** 133° **D.** 137°

15. Which angle has the same reference angle as 112°?

A. 22° **B.** 248° **C.** 202° **D.** 158°

16. The terminal arm of an angle θ passes through the point $(-7, -2)$. Which statement is true?

A. θ terminates in quadrant 2

B. $\theta_R = \theta - 180°$

C. $270° < \theta < 360°$

D. $\theta_R = 180° - \theta$

17. Numerical Response The largest angle from 0° to 360° with the same reference angle as 158° is _____°.

18. Numerical Response The point $(-4, -4)$ is on the terminal arm of θ. The value of θ is _____°.

Lesson 2: Trigonometric Ratios from a Point on the Terminal Arm

Assignment

1. Determine the exact distance r from the origin to each point. Write radicals in simplest form.
(a) $(-7, 24)$ (b) $(9, -12)$ (c) $(-2, -10)$

(d) $(5, 5)$ (e) $(-1, 3)$ (f) $(0, -4)$

2. The point $P(x, y)$ is on the terminal arm of θ, and r is the distance from the origin to P.
(a) Write $\sin\theta$, $\cos\theta$ and $\tan\theta$ in terms of x, y and r. $\sin\theta =$ _____ $\cos\theta =$ _____ $\tan\theta =$ _____
(b) Why is r never negative, even when x and y are both negative?

3. The point $(40, 9)$ is on the terminal arm of θ. Determine r and the exact values of the three primary trigonometric ratios.

4. Each point is on the terminal arm of θ. Determine the exact values of $\sin\theta$, $\cos\theta$ and $\tan\theta$ (rational denominators).
(a) $(-21, 20)$ (b) $(12, -35)$ (c) $(-3, -6)$

(d) $(2, -3)$ (e) $(-1, 1)$

5. Each point is on the terminal arm of a quadrantal angle θ. State θ and the values of $\sin\theta$, $\cos\theta$ and $\tan\theta$.
(a) $(-3, 0)$ (b) $(0, 2.5)$

6. For which angle(s) from $0°$ to $360°$ is each statement true?
(a) $\tan\theta$ is undefined _____ (b) $\sin\theta = 0$ _____ (c) $\cos\theta = -1$ _____

7. Without calculating, decide whether each ratio is positive or negative. Explain using the signs of x and y.
(a) $\sin\theta$, for the point $(-5, 2)$ (b) $\cos\theta$, for the point $(-5, 2)$

(c) $\tan\theta$, for the point $(-3, -8)$ (d) $\cos\theta$, for the point $(4, -1)$

8. The point $A(5, 12)$ is on the terminal arm of an angle α, and its reflection in the y-axis, $B(-5, 12)$, is on the terminal arm of an angle β.
(a) Determine the exact values of the three ratios for α and for β.

(b) Which ratios are equal? Which are opposites? Explain why using the reflection.

(c) Determine α and β to the nearest tenth of a degree.

9. Write each ratio as the same ratio of its reference angle, with the correct sign.
(a) $\sin 215°$ (b) $\cos 110°$ (c) $\tan 290°$

(d) $\cos 345°$ (e) $\tan 200°$ (f) $\sin 99°$

10. Each point is on the terminal arm of θ, $0° \le \theta < 360°$. Determine θ to the nearest tenth of a degree.
(a) $(7, 3)$ (b) $(-4, 9)$ (c) $(-5, -8)$ (d) $(6, -11)$ (e) $(0, -6)$

11. To find θ for the point $P(-2, 5)$, Sam entered $\tan^{-1}\left(\frac{5}{-2}\right)$ and wrote $\theta = -68.2°$. Explain why this is not the answer and find the correct θ to the nearest tenth.

12. A surveyor sets up a coordinate grid with the origin at a survey marker, the positive x-axis pointing east and the positive y-axis pointing north (units: metres). A tree stands at $(-18, 24)$.

(a) Sketch the situation and determine the distance from the marker to the tree.

(b) Determine the exact values of $\sin\theta$ and $\cos\theta$, where θ is the rotation angle from east to the tree.

(c) Determine θ to the nearest tenth of a degree.

13. The point $(4, -7)$ is on the terminal arm of an angle A. The exact value of $\cos A$ is

A. $\frac{4\sqrt{65}}{65}$ **B.** $-\frac{7\sqrt{65}}{65}$ **C.** $\frac{4\sqrt{33}}{33}$ **D.** $-\frac{4}{7}$

14. The point $(0, -3)$ is on the terminal arm of θ. Which statement is true?

A. $\sin\theta = 0$ **B.** $\cos\theta = -1$ **C.** $\tan\theta$ is undefined **D.** $\sin\theta = 1$

15. For which point on the terminal arm is $\tan\theta$ positive and $\cos\theta$ negative?

A. $(3, 5)$ **B.** $(-3, 5)$ **C.** $(-3, -5)$ **D.** $(3, -5)$

16. Numerical Response The point $(-20, -21)$ is on the terminal arm of θ, $0° \leq \theta < 360°$. To the nearest degree, $\theta =$ _____°.

17. Numerical Response The point $(-8, -6)$ is on the terminal arm of θ. To the nearest hundredth, $\tan\theta =$ _____.

Lesson 3: The CAST Rule: Signs and Solving for Angles

Assignment

1. In which quadrant(s) does the terminal arm of θ lie if:

(a) $\sin\theta > 0$? _____ (b) $\cos\theta < 0$? _____ (c) $\tan\theta > 0$? _____

(d) $\sin\theta < 0$ and $\cos\theta < 0$? _____ (e) $\sin\theta > 0$ and $\tan\theta < 0$? _____ (f) $\cos\theta > 0$ and $\sin\theta < 0$? _____

2. Without using a calculator, state whether each ratio is positive or negative.

(a) $\sin 203°$ _____ (b) $\cos 147°$ _____ (c) $\tan 112°$ _____

(d) $\cos 289°$ _____ (e) $\tan 199°$ _____ (f) $\sin 138°$ _____

3. Write each ratio as the same ratio of its reference angle, with the correct sign.

(a) $\cos 168°$ _____ (b) $\tan 236°$ _____ (c) $\sin 302°$ _____

(d) $\cos 251°$ _____ (e) $\sin 117°$ _____ (f) $\tan 341°$ _____

4. Determine the other angle θ, $0° \leq \theta < 360°$, that satisfies each equation.

(a) $\sin\theta = \sin 52°$ (b) $\cos\theta = \cos 233°$ (c) $\tan\theta = \tan 161°$

5. Determine the exact values of the other two ratios. Sketch each angle.

(a) $\cos\theta = \frac{20}{29}$, $270° < \theta < 360°$ (b) $\tan A = -\frac{12}{35}$, $0° \leq A \leq 180°$ (c) $\cos\theta = 0.28$ and $\tan\theta < 0$

6. Given $\sin\theta = \frac{5}{6}$ and $\cos\theta < 0$, determine the exact values of $\cos\theta$ and $\tan\theta$ with rational denominators.

7. Given $\cos\theta = -\frac{\sqrt{5}}{4}$ and $\tan\theta > 0$, determine the exact values of $\sin\theta$ and $\tan\theta$ with rational denominators.

8. Solve for θ, $0° \leq \theta < 360°$, to the nearest tenth of a degree.
(a) $\sin\theta = 0.8$ (b) $\cos\theta = 0.35$ (c) $\tan\theta = 1.9$

(d) $\sin\theta = -0.27$ (e) $\cos\theta = -0.91$ (f) $\tan\theta = -0.6$

9. Solve for θ, $0° \leq \theta < 360°$. Round to the nearest tenth of a degree where needed.
(a) $\cos\theta = 1$ (b) $\sin\theta = 0$ (c) $\cos\theta = -1$ (d) $\sin\theta = -1.3$ (e) $\tan\theta = -40$

10. Solve for θ, $0° \leq \theta < 360°$, to the nearest tenth of a degree.
(a) $4\sin\theta - 1 = 0$ (b) $5\cos\theta + 3 = 1$

(c) $2\tan\theta + 9 = 4$ (d) $3\sin\theta + 7 = 2$

11. Solve each equation in the given interval, to the nearest tenth of a degree.
(a) $\cos\theta = -0.15$, $0° \leq \theta \leq 180°$ (b) $\sin\theta = 0.55$, $90° \leq \theta \leq 270°$ (c) $\tan\theta = 0.7$, $180° \leq \theta < 360°$

12. A robot arm 1.2 m long pivots about a fixed point. When the arm has rotated θ in standard position, its tip is $x = 1.2\cos\theta$ metres right of the pivot and $y = 1.2\sin\theta$ metres above it (negative values mean left or below). Answer to the nearest tenth of a degree, $0° \leq \theta < 360°$.

(a) At what angles is the tip 0.5 m below the pivot?

(b) At what angles is the tip 0.9 m left of the pivot?

13. Jordan solved $\cos\theta = -0.6$ for $0° \le \theta < 360°$ this way: "$\cos^{-1}(-0.6) \approx 126.9°$. Cosine is negative in Quadrants II and IV, so $\theta \approx 126.9°$ and $180° + 126.9° = 306.9°$." Identify the errors and give the correct solutions.

14. Which ratio has a different sign from the other three?

A. $\sin 140°$ **B.** $\cos 320°$ **C.** $\tan 230°$ **D.** $\cos 200°$

15. If $\sin\theta = -\frac{9}{41}$ and $180° < \theta < 270°$, then the values of $\cos\theta$ and $\tan\theta$ are

A. $-\frac{40}{41}$ and $\frac{9}{40}$ **B.** $\frac{40}{41}$ and $-\frac{9}{40}$ **C.** $-\frac{40}{41}$ and $-\frac{9}{40}$ **D.** $-\frac{40}{41}$ and $\frac{40}{9}$

16. To the nearest degree, the solutions of $\cos\theta = 0.24$ for $0° \le \theta < 360°$ are

A. $76°, 104°$ **B.** $76°, 284°$ **C.** $104°, 256°$ **D.** $76°, 256°$

17. Numerical Response Angles A and B are between $0°$ and $360°$, with $\cos A = \cos 314°$ ($A \ne 314°$) and $\sin B = \sin 251°$ ($B \ne 251°$). The value of $A + B$, in degrees, is _____.

18. Numerical Response For $0° \le \theta \le 180°$, the solution of $9\cos\theta + 4 = 0$, to the nearest degree, is _____°.

Lesson 4: Exact Values from Special Triangles

Assignment

1. From memory, sketch the two special triangles with all side lengths and angles. Use them to write each exact value in rationalized form.

(a) $\cos 30°$ (b) $\tan 45°$ (c) $\sin 60°$ (d) $\tan 30°$ (e) $\cos 60°$ (f) $\sin 45°$

_____ _____ _____ _____ _____ _____

2. Determine the exact value.

(a) $\cos 150°$ (b) $\sin 225°$ (c) $\tan 315°$ (d) $\sin 120°$ (e) $\cos 240°$ (f) $\tan 210°$

_____ _____ _____ _____ _____ _____

(g) $\cos 90°$ (h) $\tan 180°$ (i) $\sin 270°$ (j) $\tan 90°$ (k) $\tan 135°$ (l) $\sin 210°$

_____ _____ _____ _____ _____ _____

3. Patterns in the exact values.

(a) Use the 30-60-90 triangle to explain why $\sin 30° = \cos 60°$.

(b) Show that $\tan\theta = \dfrac{\sin\theta}{\cos\theta}$ for $\theta = 60°$ and for $\theta = 225°$.

(c) As θ increases from $0°$ to $90°$, describe what happens to $\sin\theta$, $\cos\theta$ and $\tan\theta$.

4. Determine the exact value of each expression.

(a) $\cos 60° + \sin 30°$ (b) $\tan^2 300° - 1$ (c) $\sin 135° \cos 225°$

(d) $4 \sin 210° \cos 330°$ (e) $\dfrac{1 - \cos 120°}{\sin 270°}$ (f) $\sin^2 240° + \cos^2 240°$

5. True or false? Justify with exact values.

(a) $\cos 120^\circ = -\cos 60^\circ$

(b) $\cos(2 \times 45^\circ) = 2\cos 45^\circ$

(c) $\tan 225^\circ = \tan 45^\circ$

(d) $\sin 300^\circ = \sin 60^\circ$

6. Solve for θ, $0^\circ \leq \theta < 360^\circ$, using exact values.

(a) $\cos\theta = \dfrac{1}{2}$

(b) $\sin\theta = -\dfrac{\sqrt{2}}{2}$

(c) $\tan\theta = -1$

(d) $\sin\theta = \dfrac{\sqrt{3}}{2}$

(e) $\cos\theta = -1$

(f) $\tan\theta = -\sqrt{3}$

(g) $\sin\theta = 0$

(h) $\tan\theta$ is undefined

7. Solve for θ, $0^\circ \leq \theta < 360^\circ$, using exact values.

(a) $2\cos\theta + \sqrt{3} = 0$

(b) $\sqrt{2}\cos\theta - 1 = 0$

(c) $3\tan\theta - \sqrt{3} = 0$

(d) $5 - 2\sin\theta = 4$

8. Solve for θ, $0^\circ \leq \theta < 360^\circ$, using exact values.

(a) $4\cos^2\theta = 1$

(b) $2\sin^2\theta - 1 = 0$

(c) $3\tan^2\theta = 1$

9. Solve each equation in the given interval, using exact values.

(a) $\cos\theta = -\dfrac{\sqrt{3}}{2}$, $180° \le \theta < 360°$ (b) $\tan\theta = \sqrt{3}$, $90° \le \theta \le 270°$ (c) $2\sin\theta = 1$, $90° \le \theta \le 270°$

10. Each point is on the terminal arm of θ. Determine θ exactly, $0° \le \theta < 360°$.
(a) $(-5, 5\sqrt{3})$ (b) $(6, -6)$ (c) $(-2\sqrt{3}, -2)$

11. Determine the angle ($0° \le \theta < 360°$) and the exact values of the other two ratios.
(a) $\cos A = \dfrac{\sqrt{2}}{2}$ and A is not in Quadrant I (b) $\tan B = -\sqrt{3}$ and $\sin B > 0$

12. A point moves counterclockwise around a circle centred at the origin, from $P(2, 2\sqrt{3})$ to $Q(2\sqrt{3}, -2)$. Through what angle does it rotate?

13. Sam wrote: "$\sin 240° = \dfrac{\sqrt{3}}{2}$ because the reference angle is 60°" and "$\tan 150° = -\sqrt{3}$ because 150° is in Quadrant II." Find and correct each error.

14. The exact value of $\cos 210° + \sin 300°$ is
A. $-\sqrt{3}$ **B.** 0 **C.** $\sqrt{3}$ **D.** $-\dfrac{1+\sqrt{3}}{2}$

15. The smallest positive solution of $2\cos\theta + \sqrt{2} = 0$ is
A. 45° **B.** 135° **C.** 225° **D.** 315°

16. Which of the following is **not** equal to $\dfrac{1}{2}$?
A. $\sin 150°$ **B.** $\cos 300°$ **C.** $-\sin 210°$ **D.** $\cos 120°$

17. Numerical Response The sum of all the solutions of $2\cos\theta - \sqrt{3} = 0$ for $0° \leq \theta < 360°$, in degrees, is _____.

18. Numerical Response The exact value of $(\tan 60°)(\tan 240°) + 4\cos^2 45°$ is _____.

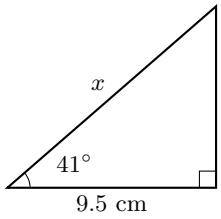
Lesson 5: The Sine Law

Assignment

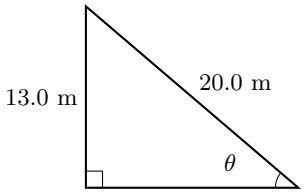
Round sides to the nearest tenth and angles to the nearest degree unless a question says otherwise. Keep unrounded values in your calculator until the final answer.

1. Determine the indicated value in each right triangle.

(a)



(b)

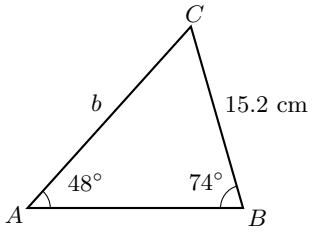


2. (a) Write the sine law for $\triangle KLM$.

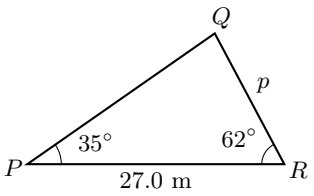
(b) In $\triangle RST$, which side is opposite $\angle S$? Which angle is opposite side RS?

3. Determine the indicated side.

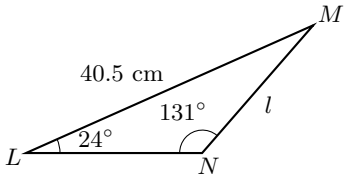
(a)



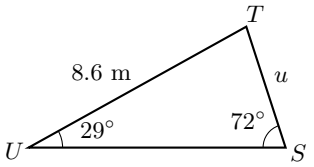
(b)



(c)

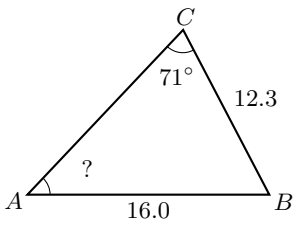


(d)

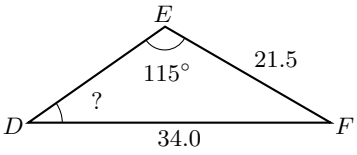


4. Determine the indicated angle. Each triangle has only one possible shape: state the reason.

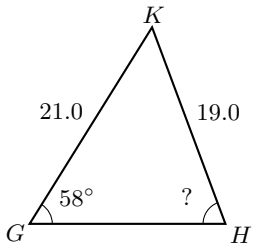
(a)



(b)



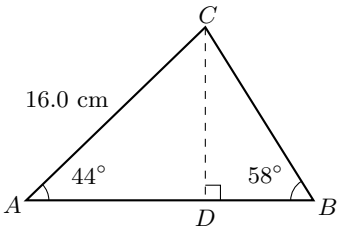
(c) The triangle is acute.



5. In $\triangle ABC$, $\angle A = 53°$, $\angle C = 68°$ and $c = 21.4$ cm. Sketch the triangle, then determine a and b.

6. In $\triangle PQR$, $\angle Q = 102°$, $q = 45$ m and $r = 27$ m. Explain why $\angle R$ must be acute, then solve the triangle.

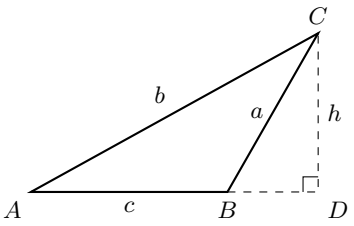
7. In $\triangle ABC$, $\angle A = 44°$, $\angle B = 58°$ and $AC = 16.0$ cm. The altitude from C meets AB at D.



(a) Without the sine law, determine CD and then BC.

(b) Check your answer with the sine law.

8. In $\triangle ABC$ below, $\angle B$ is obtuse, so the altitude CD meets side AB extended.

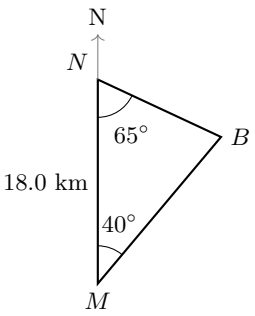


(a) Explain why $\angle CBD = 180° - \angle B$.

(b) Write h in terms of b and $\angle A$, and in terms of a and $\angle B$.

(c) Show that $\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$ in this triangle too.

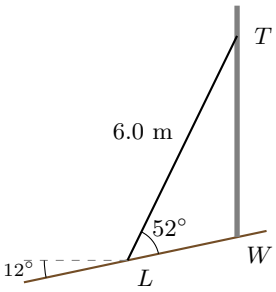
9. Coast guard station N is 18.0 km due north of station M. A boat B is on a bearing of N40°E from M and S65°E from N.



(a) Determine the three angles of $\triangle MNB$.

(b) Which station is closer to the boat, and by how much?

10. A 6.0 m ladder leans against a vertical wall. The ground slopes up toward the wall at 12° to the horizontal, and the ladder makes an angle of 52° with the sloped ground.



(a) Explain why the angle between the wall and the ground at the foot of the wall, $\angle LWT$, is 102°.

(b) How far up the wall, measured from W, does the ladder reach?

11. A triangular field has corners J, K and L. The side JK is 185 m, $\angle J = 63°$ and $\angle K = 49°$.

(a) Sketch the field and determine its perimeter to the nearest metre.

(b) Determine the area of the field to the nearest ten square metres. (Hint: find the height from L to JK.)

12. In $\triangle JKL$, $\angle J = 25°$, $\angle L = 140°$ and $KL = 7.5$ m. The length of JK, to the nearest tenth of a metre, is

A. 4.6 m **B.** 4.9 m **C.** 11.4 m **D.** 18.6 m

13. In acute $\triangle ABC$, $\angle A = 40°$, $a = 11$ cm and $c = 14$ cm. To the nearest tenth of a degree, $\angle C$ is

A. 30.3° **B.** 54.9° **C.** 85.1° **D.** 125.1°

14. For which set of given information can the sine law *not* be used as the first step?

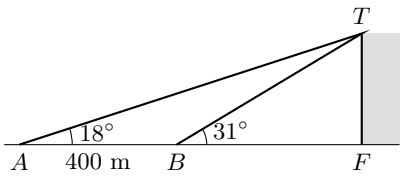
A. $\angle A$, $\angle B$ and a

B. $\angle B$, $\angle C$ and a

C. a, b and $\angle C$

D. a, b and $\angle A$ $(a > b)$

15. Numerical Response From a boat at point A, the angle of elevation of the top T of a vertical cliff is 18°. The boat moves 400 m straight toward the cliff to point B, where the angle of elevation is 31°. The height of the cliff, to the nearest metre, is _____ m.



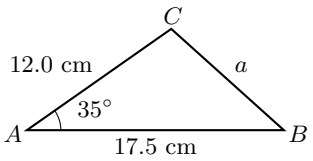
Lesson 6: The Cosine Law

Assignment

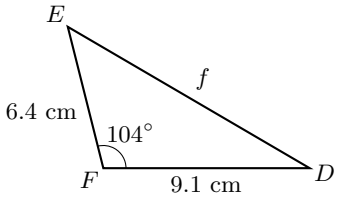
Round sides to the nearest tenth and angles to the nearest degree unless a question says otherwise.

1. For $\triangle KLM$, complete each form of the cosine law.
(a) $k^2 =$ _____ (b) $m^2 =$ _____ (c) $\cos L =$ _____

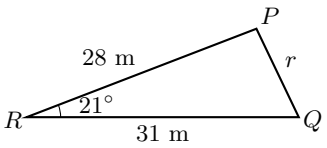
2. Determine the indicated side.
(a)



(b)



(c)



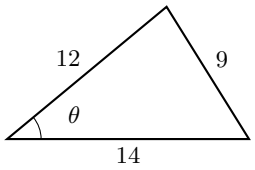
3. In $\triangle ABC$, $\angle B = 76°$, $a = 23$ and $c = 31$. Determine b to the nearest whole number.

4. Without a calculator, determine p exactly in $\triangle PQR$, in simplest radical form.
(a) $\angle P = 60°$, $q = 6$, $r = 10$

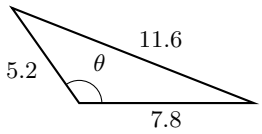
(b) $\angle P = 135°$, $q = 2$, $r = 3\sqrt{2}$

5. Determine the indicated angle.

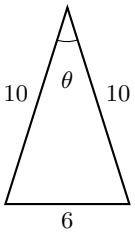
(a)



(b)



(c)



6. Three trees in a garden are 17 m, 25 m and 30 m apart. Determine the smallest and the largest angle of the triangle they form.

7. Solve $\triangle ABC$ if $\angle C = 112°$, $a = 8.4$ m and $b = 13.6$ m.

8. In $\triangle ABC$, $\angle A = 53°$, $AC = 10.0$ cm and $AB = 16.0$ cm.

(a) Draw the altitude from C to AB and use right triangles to determine BC.

(b) Check your answer with the cosine law.

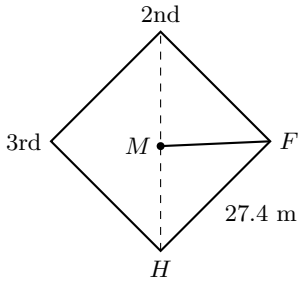
9. (a) What does the cosine law $a^2 = b^2 + c^2 - 2bc\cos A$ become when $\angle A = 90°$? Why?

(b) Explain why $a^2 > b^2 + c^2$ when $\angle A$ is obtuse.

10. Two hikers leave a trailhead at the same time on straight paths that make an angle of 68°. One walks at 4.5 km/h and the other at 3.8 km/h. How far apart are they after 2.5 h?

11. A triangular lot has sides of 42 m, 57 m and 68 m. Determine all three angles. Check that they add to 180°.

12. A baseball diamond is a square with sides of 27.4 m. The pitcher's mound M is 18.4 m from home plate H, on the diagonal from home plate to second base. Determine the distance from the mound to first base F.



13. A parallelogram has sides of 7.0 cm and 11.0 cm, and its shorter diagonal is 8.0 cm. Determine the angles of the parallelogram and the length of its longer diagonal.

14. A student tries to find the largest angle of a triangle with sides 4, 7 and 12. Show what happens and explain why.

15. The sides of a triangle are in the ratio $3 : 5 : 7$. Determine the exact measure of its largest angle.

16. In $\triangle ABC$, $AB = 4$, $AC = 6$ and $\angle A = 150°$. The exact value of $(BC)^2$ is

A. $52 - 24\sqrt{3}$ **B.** $52 + 24\sqrt{3}$ **C.** $52 + 48\sqrt{3}$ **D.** 28

17. In $\triangle ABC$, $b = 5$, $c = 8$ and $\angle A = 50°$. To the nearest tenth, a is

A. 2.4 **B.** 6.1 **C.** 11.9 **D.** 37.6

18. Numerical Response Two ships leave port P. One sails 9.5 km to R and the other sails 11.0 km to Q. The ships are then 16.5 km apart. The angle between their directions, $\angle RPQ$, to the nearest degree, is _____°.

Lesson 7: The Ambiguous Case

Assignment

1. Explain what the ambiguous case is. Describe, with a sketch, when SSA information gives no triangle, one triangle and two triangles.

2. Name the type of information (SSS, SAS, ASA, AAS or SSA). State whether exactly one triangle is guaranteed.

(a) $\triangle DEF$: $d = 5$, $e = 9$, $\angle F = 64°$

(b) $\triangle DEF$: $\angle D = 33°$, $d = 6$, $f = 10$

(c) $\triangle STU$: $\angle S = 41°$, $\angle T = 75°$, $s = 12$

(d) $\triangle STU$: $s = 4$, $t = 7$, $u = 10$

3. In $\triangle ABC$, $\angle B = 42°$ and $c = AB = 15$ cm. Side $b = AC$ is opposite $\angle B$.

(a) Determine the height h from A to line BC, to the nearest tenth.

(b) How many triangles are possible if $b = 8$ cm? $b = 12$ cm? $b = 18$ cm?

4. In $\triangle ABC$, $\angle A = 36°$ and $b = 20$ cm. For what lengths of a (to the nearest tenth) is there (a) no triangle, (b) exactly one triangle, (c) two triangles?

5. Sketch each triangle. Without using the sine law, decide whether there are 0, 1 or 2 triangles.

(a) $\triangle DEF$: $\angle D = 64°$, $d = 10.2$ cm, $e = 8.5$ cm

(b) $\triangle STV$: $\angle S = 33°$, $s = 5.4$ m, $t = 9.0$ m

(c) $\triangle JKL$: $\angle J = 118°$, $j = 7$ mm, $k = 9$ mm

(d) $\triangle MNP$: $\angle M = 55°$, $m = 6$ cm, $n = 8$ cm

6. Determine all possible measures of $\angle C$, to the nearest degree.

(a) $\triangle ABC$: $\angle A = 34°$, $a = 6.2$ cm, $c = 9.5$ cm

(b) $\triangle ABC$: $\angle A = 71°$, $a = 12.4$ cm, $c = 9.8$ cm

7. In $\triangle LMN$, $\angle N = 39°$, $LM = 26$ mm and $LN = 35$ mm. Determine all possible measures of $\angle M$ and of $\angle L$, to the nearest degree.

8. Solve all possible triangles ABC with $\angle B = 48°$, $b = 11$ cm and $a = 14$ cm. Round angles to the nearest degree and sides to the nearest tenth.

9. In $\triangle XYZ$, $\angle X = 124°$ and $y = 12$ cm.

(a) If $x = 19$ cm, solve the triangle (angles to the nearest degree, sides to the nearest tenth).

(b) If $x = 10$ cm, a student gets $\sin Y \approx 0.9948$ and says $Y \approx 84°$. Explain the error.

10. Jordan solved $\triangle ABC$ with $\angle A = 30°$, $a = 8$ cm and $b = 11$ cm and wrote "$\angle B = 43°$". What did Jordan miss? Give the complete answer to the nearest degree.

11. A lighthouse L is 14 km from a harbour H. A boat leaves H on a straight course that makes an angle of 25° with the line HL. The lighthouse foghorn can be heard up to 8 km away. Sketch the situation. Between what distances from

H (to the nearest tenth of a km) can the boat hear the foghorn?

12. A garden gate 3.0 m wide is hinged at post G. Post G is 4.0 m from corner K of a straight path, and line KG makes an angle of 35° with the edge of the path. The gate swings until its free end touches the edge of the path. Find both possible distances from K to the point where the gate touches the path, to the nearest tenth of a metre.

13. In $\triangle ABC$, $\angle A = 30°$ and $b = 12$ cm. List every whole-number length of a (in cm) that gives two triangles.

14. In $\triangle ABC$, $\angle A = 40°$, $a = 8$ cm and $b = 10$ cm. Vertex B can be in two places on the same line. How far apart are the two possible positions of B? Answer to the nearest tenth.

15. In $\triangle PQR$, $\angle P = 46°$, $QR = 9$ cm and $PR = 12$ cm. To the nearest tenth of a degree, the measure of $\angle Q$ is

A. 73.6°

B. 73.6° or 106.4°

C. 32.6°

D. 32.6° or 147.4°

16. Which set of measurements gives exactly two triangles ABC?

A. $\angle A = 50°$, $a = 12$ cm, $b = 10$ cm

B. $\angle A = 50°$, $a = 7$ cm, $b = 10$ cm

C. $\angle A = 50°$, $a = 9$ cm, $b = 10$ cm

D. $\angle A = 130°$, $a = 9$ cm, $b = 10$ cm

17. Numerical Response In $\triangle ABC$, $\angle A = 33°$, $a = 10$ cm and $b = 16$ cm. To the nearest degree, the obtuse possible measure of $\angle B$ is _____°.

Lesson 8: Choosing the Sine Law or the Cosine Law

Assignment

1. Name the most appropriate first method and justify it. Do not solve.

(a) $\triangle ABC$: $a = 12$, $b = 15$, $c = 20$; find $\angle C$

(b) $\triangle DEF$: $\angle E = 90°$, $e = 17$, $d = 8$; find $\angle D$

(c) $\triangle GHK$: $\angle G = 63°$, $\angle H = 48°$, $h = 14$; find g

(d) $\triangle LMN$: $l = 10$, $\angle N = 35°$; find n

2. The minute hand of a clock is 15 cm long and the hour hand is 11 cm long. Round distances to the nearest tenth.

(a) Find the angle between the hands at 2:00 and the distance between the tips.

(b) Find the angle between the hands at 4:30 and the distance between the tips.

(c) What is the distance between the tips at 6:00?

3. Write each bearing in the other form.

(a) 035°

(b) 160°

(c) 215°

(d) 290°

(e) S 25° E

(f) N 15° W

(g) S 80° W

(h) N 5° E

(i) Town B is on a bearing of 035° from town A. What is the bearing of A from B? _____

4. A plane flies 150 km on a bearing of 300° from base P to Q, then 210 km on a bearing of 050° to R. Find the distance from R back to P (nearest kilometre) and the bearing it must fly to return (nearest degree).

5. At noon a ship at A sees a lighthouse L on a bearing of 025°. The ship sails 15 km on a bearing of 095° to B, where the lighthouse is on a bearing of 330°.

(a) Sketch the situation and find the three angles of $\triangle ABL$.

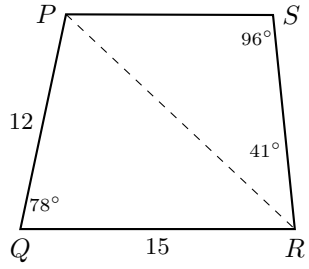
(b) Find the distance BL, to the nearest tenth of a kilometre.

(c) How close to the lighthouse did the ship pass on its way from A to B?

6. Two lamp posts stand 80 m apart on level ground. From point P on the ground between them, 30 m from the first post, the angle of elevation to the top of the first post is 55° and to the top of the second post is 15°. Find the distance between the tops of the posts, to the nearest tenth of a metre.

7. A hot-air balloon H is directly above a point between two observers A and B, who are 600 m apart. The angles of elevation of the balloon from A and B are 40° and 56°. Find the height of the balloon, to the nearest metre.

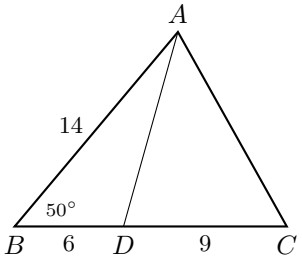
8. Quadrilateral $PQRS$ is split by diagonal PR. $PQ = 12$ cm, $QR = 15$ cm, $\angle PQR = 78°$, $\angle PRS = 41°$ and $\angle PSR = 96°$.



(a) Find PR, PS and RS to the nearest tenth.

(b) Find the perimeter and the area of $PQRS$, to the nearest tenth.

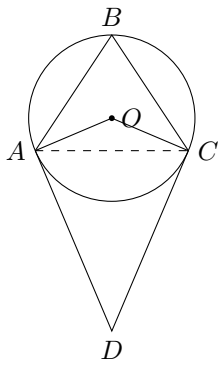
9. In $\triangle ABC$, $AB = 14$ cm and $\angle ABC = 50°$. Point D is on BC with $BD = 6$ cm and $DC = 9$ cm.



Find, to the nearest tenth or nearest degree:
(a) AD (b) $\angle ADB$ (c) AC (d) $\angle ACB$

10. A circle with centre O has radius 9 cm. Chord PQ is 14 cm long, and R is a point on the major arc. Find $\angle POQ$ and $\angle PRQ$ to the nearest degree.

11. Tangents DA and DC touch a circle with centre O and radius 5 cm at A and C. $\angle ADC = 46°$, and B is a point on the major arc AC.



(a) Explain why $\angle AOC = 134°$.
(b) Find the length of chord AC to the nearest tenth.
(c) Find $\angle ABC$.
(d) Find the length of the tangent DA to the nearest tenth.

12. Find each area, to the nearest tenth.
(a) $\triangle ABC$ with $a = 11$ cm, $b = 16$ cm and $\angle C = 38°$.

(b) A triangle with sides 8 m, 10 m and 13 m.

13. A triangle has sides of 12 cm and 15 cm and an area of 60 cm^2. Find both possible measures of the angle between the two sides, to the nearest tenth of a degree. Why are there two?

14. A sailboat leaves a dock, sails 6 km on a bearing of 120°, then turns onto a bearing of 030° and sails until it is due east of the dock. To find how far it is from the dock, the most appropriate method is

A. primary trigonometric ratios

B. the sine law

C. the cosine law

D. the problem cannot be solved without more information

15. In $\triangle PQR$, $PQ = 7$ cm, $PR = 11$ cm and $\angle PQR = 105°$. To find $\angle PRQ$, the most appropriate method is

A. primary trigonometric ratios

B. the sine law

C. the cosine law

D. the problem cannot be solved without more information

16. The area of a triangle with sides 10 cm and 14 cm and a contained angle of 150° is

A. 35.0 cm^2 **B.** 60.6 cm^2 **C.** 70.0 cm^2 **D.** 121.2 cm^2

17. Numerical Response To the nearest degree, the smaller angle between the hands of a clock at 7:20 is _____°.

18. Numerical Response Two hikers leave camp at the same time. One walks on a bearing of 020° at 4.5 km/h and the other on a bearing of 130° at 5.0 km/h. To the nearest tenth of a kilometre, their distance apart after 2 hours is _____ km.

Lesson 9: Unit 3 Review

Assignment

Use Degree mode. Sides to the nearest tenth and angles to the nearest degree unless a question says otherwise. Questions 4 and 5 ask for exact answers.

1. (a) Complete the table.

θ	112°	196°	333°	87°
Quadrant				
θ_R				

(b) Determine all angles from 0° to 360° with a reference angle of 52°. _____

(c) Angle θ terminates in quadrant III and $\theta_R = 41°$. Determine θ. _____

2. Each point is on the terminal arm of an angle θ in standard position. Determine r, the exact values of the three primary ratios, and θ to the nearest tenth of a degree.

(a) $P(-20, 21)$ (b) $P(-1, -3)$ (c) $P(0, -4)$

3. Determine the exact values of the other two primary ratios.

(a) $\cos\theta = \frac{3}{8}$ and $\sin\theta < 0$ (b) $\tan\theta = -\frac{4}{7}$ and $90° < \theta < 180°$

4. Determine the exact value.

(a) $\cos 240°$ (b) $\tan 315°$ (c) $\sin 120°$ (d) $\cos 330°$

(e) $\tan 150°$ (f) $\sin 225°$ (g) $\cos 180°$ (h) $\tan 90°$

(i) $\sin 30° \cos 60° + \tan 135°$ (j) $\cos^2 315° - \sin^2 240°$ (k) $\dfrac{\sin 270° + \cos 120°}{\tan 225°}$

5. Solve exactly for $0° \leq \theta < 360°$.

(a) $\sin\theta = \dfrac{\sqrt{3}}{2}$ (b) $2\cos\theta + 1 = 0$ (c) $\sqrt{3}\tan\theta = -1$ (d) $\cos\theta = 0$

6. Solve for $0° \leq \theta < 360°$, to the nearest tenth of a degree.

(a) $\cos\theta = 0.215$ (b) $\sin\theta = -0.64$ (c) $5\tan\theta - 8 = 0$ (d) $4\sin\theta = 5$

7. Each student made an error. Explain it and give the correct answer.

(a) Jordan wrote: "The reference angle of 230° is 130°."

(b) Mei solved $\cos\theta = -0.3$ for $0° \leq \theta < 360°$. She entered $\cos^{-1}(-0.3) \approx 107.5°$, then added 180° and wrote $\theta = 107.5°, 287.5°$.

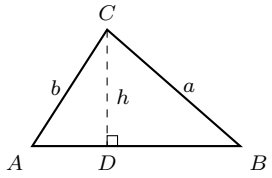
8. (a) In $\triangle DEF$, $\angle D = 54°$, $\angle E = 67°$ and $e = 18.5$ cm. Determine d and f.

(b) In $\triangle XYZ$, $\angle X = 112°$, $x = 11.4$ m and $z = 8.7$ m. Determine $\angle Z$ and $\angle Y$.

9. (a) In $\triangle ABC$, $\angle B = 37°$, $a = 14$ cm and $c = 22$ cm. Determine b.

(b) A triangle has sides 8 cm, 11 cm and 16 cm. Determine all three angles.

10. (a) Complete the reasons in this proof of the sine law for acute $\triangle ABC$, with altitude $h = CD$ from C to AB.



$\sin A = \dfrac{h}{b}$, so $h = b \sin A$ _____

$\sin B = \dfrac{h}{a}$, so $h = a \sin B$ _____

$a \sin B = b \sin A$ _____

$\dfrac{a}{\sin A} = \dfrac{b}{\sin B}$ _____

(b) In $\triangle ABC$, $\angle A = 50°$, $\angle B = 60°$ and $b = 10$ cm. Without using the sine law, determine a by drawing the altitude from C and using primary ratios. Check with the sine law.

11. For each $\triangle ABC$, state whether there are 0, 1 or 2 triangles. Show the comparison you used.

(a) $\angle A = 40°$, $a = 5$, $b = 12$ (b) $\angle A = 30°$, $a = 6$, $b = 12$ (c) $\angle A = 52°$, $a = 10$, $b = 11$

(d) $\angle A = 115°$, $a = 9$, $b = 7$ (e) $\angle A = 115°$, $a = 7$, $b = 9$ (f) $\angle A = 65°$, $a = 14$, $b = 9$

12. In $\triangle ABC$, $\angle A = 35°$, $a = 8.2$ cm and $b = 12.5$ cm. Show that there are two triangles and solve both.

13. A ship leaves port and sails 28 km on a bearing of 062°. It then turns and sails 19 km on a bearing of 154°.

(a) Sketch a diagram and show that the angle at the turning point is 88°.

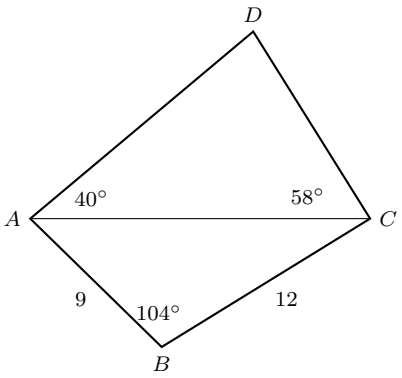
(b) How far is the ship from port?

(c) On what bearing must the ship sail to return directly to port? (Find the angle at port first.)

14. In quadrilateral $ABCD$, $AB = 9$ m, $BC = 12$ m, $\angle ABC = 104°$, $\angle CAD = 40°$ and $\angle ACD = 58°$.

(a) Determine the length of the diagonal AC. Justify your choice of law.

(b) Determine AD and CD.



(c) Determine the perimeter and the area of $ABCD$, using Area $= \frac{1}{2}ab\sin C$ for each triangle.

15. Two straight roads meet at junction J at an angle of 34°. A cell tower T is on the first road, 6.0 km from J. Its signal reaches 4.0 km in every direction.

(a) Sketch the situation. Explain why a car on the second road has signal for only part of the road.

(b) Determine, to the nearest tenth of a kilometre, the distances from J along the second road at which the car gains and loses the signal.

16. Which pair of angles in standard position have the same reference angle?
A. 40° and 130° **B.** 125° and 235° **C.** 200° and 290° **D.** 75° and 165°

17. The point (a, b), with $a > 0$ and $b > 0$, is on the terminal arm of angle θ. The point $(-a, b)$ is on the terminal arm of
A. $180° - \theta$ **B.** $180° + \theta$ **C.** $360° - \theta$ **D.** $90° + \theta$

18. Numerical Response The terminal arm of angle θ passes through $(-9, 40)$. To the nearest degree, θ is _____°.

19. In which quadrant are both $\cos\theta$ and $\sin\theta$ negative?
A. I **B.** II **C.** III **D.** IV

20. The exact value of $\sin 240°$ is
A. $-\frac{\sqrt{3}}{2}$ **B.** $\frac{\sqrt{3}}{2}$ **C.** $-\frac{1}{2}$ **D.** $\frac{1}{2}$

21. Numerical Response The sum of all the solutions of $2\sin\theta + 1 = 0$, $0° \le \theta < 360°$, is _____°.

22. The solutions of $\cos\theta = -0.6$, $0° \le \theta < 360°$, to the nearest tenth of a degree, are
A. $53.1°, 306.9°$ **B.** $126.9°, 233.1°$ **C.** $126.9°, 306.9°$ **D.** $233.1°, 306.9°$

23. In $\triangle ABC$, $\angle A = 44°$, $a = 11$ cm and $b = 15$ cm. The number of possible triangles is
A. 0 **B.** 1 **C.** 2 **D.** cannot be determined

24. Numerical Response Two sides of a triangular garden are 13 m and 15 m and the angle between them is 48°. The area of the garden, to the nearest tenth of a square metre, is _____ m^2.

25. A surveyor knows two sides of a triangular lot, 85 m and 62 m, and the angle of 71° between them. To find the third side, the most appropriate method is

A. SOH CAH TOA

B. the sine law

C. the cosine law

D. the problem cannot be solved without more information

26. The largest angle of a triangle with sides 6 cm, 8 cm and 12 cm is, to the nearest degree,

A. 27° **B.** 63° **C.** 117° **D.** 153°

27. Numerical Response Two hikers leave a camp at the same time. One walks 5.2 km on a bearing of 020° and the other walks 7.8 km on a bearing of 100°. To the nearest tenth of a kilometre, they are _____ km apart.
